Unscented Kalman Filter

S-88.4221 Postgraduate Seminar on Signal Processing

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Outline

- Unscented transformation (UT)
- Unscented Kalman filter (UKF)
- Example / Matlab-demo
- Homework

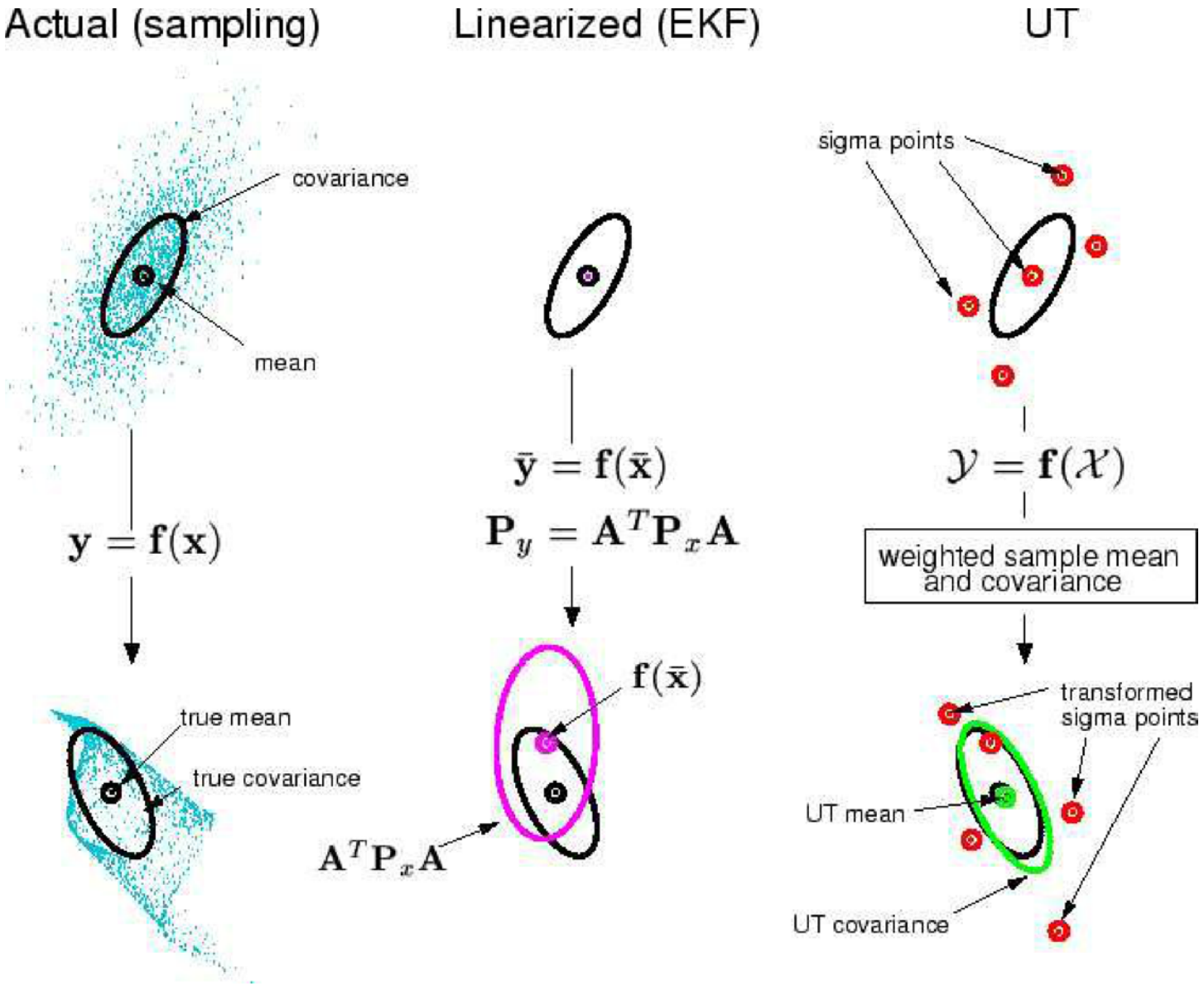
Why yet another KF

- EKF is difficult to tune, the Jacobian can be hard to derive, and it can only handle limited amount of nonlinearity
- PF can handle arbitrary distributions and non-linearities but is computationally very complex
- UKF gives a nice tradeoff between PF and EKF

Unscented transformation

- Problem: Given an n-dimensional r.v. x with mean and covariance $(\bar{x}, P_x)$, find $(\bar{y}, P_y)$ where $y = h(x)$ is a transformed r.v. and h general (non-linear) function.
- Solutions:
  - Solve analytically $\rightsquigarrow$ gray hair.
  - Use Taylor series $\rightsquigarrow$ EKF.
  - Use UT $\rightsquigarrow$ UKF.
  - (Monte-Carlo integration $\rightsquigarrow$ particle filter.)

Unscented transformation



Unscented transformation

- n-dimensional r.v. x is approximated with $2n+1$ sigma points $x^{(i)}$ and weights $W^{(i)}$.
- The sigma points are chosen such that the weighted "sample mean and covariance" of the sigma points match (x, P_x).
- Transformed sigma points yield then approximate $(\bar{y}, P_y)$ of r.v. $y = h(x)$.

Unscented transformation

$$x^{(0)} = x, \qquad w^{(0)} = \frac{\kappa}{n+\kappa}$$

$$x^{(i)} = x + \left(\sqrt{(n+\kappa)P_x}\right)_i, \qquad w^{(i)} = \frac{1}{2(n+\kappa)}$$

$$x^{(i+n)} = \bar{x} - \left(\sqrt{(n+\kappa)P_x}\right)_i, \qquad w^{(i+n)} = w^{(i)}$$

$$\bar{y} = \sum_{i=0}^{2n} W^{(i)} f(x^{(i)}), \qquad P_y = \sum_{i=0}^{2n} w^{(i)} (f(x^{(i)}) - \bar{y})(f(x^{(i)}) - \bar{y})^T \ldots$$

where $(\sqrt{A})_i$ denotes the ith column of a matrix square root of A and κ is a user-defined constant.

Unscented transformation

The mean and covariance of r.v. $y = h(x)$ can be written using Taylor series expansion of h as ($x = \bar{x} + \tilde{x}$)

$$y = h(\bar{x}) + \frac{1}{2!} E[D^2_{\tilde{x}} h] + \frac{1}{4!} E[D^4_{\tilde{x}} h] + \dots$$

$$P_y = H P_x H^T + E[\frac{1}{3!} D_{\tilde{x}} h (D^3_{\tilde{x}} h)^T + \frac{1}{2!2!} D^2_{\tilde{x}} h (D^2_{\tilde{x}} h)^T + \frac{1}{3!} D^3_{\tilde{x}} h (D_{\tilde{x}} h)^T]$$
$$+ E[\frac{1}{2!} D^2_{\tilde{x}} h] E[\frac{1}{2!} D^2_{\tilde{x}} h]^T + \dots$$

where $D^k_{\tilde{x}} h$ is shorthand notation:

$$D^k_{\tilde{x}} h = \left(\sum_{i=1}^{n} \tilde{x}_i \frac{\partial}{\partial x_i} \right)^k h(x) \Big|_{\tilde{x}}.$$

Unscented transformation

- UT gives correct mean up to third order and covariance up to the second order for any h. Linearized mean is correct only up to first order. Covariance in UT and linearization have the same order of accuracy. The magnitude of the error is, however, smaller in UT.
- UT with $\kappa \neq 0$ is termed generalized UT. With Gaussian x choosing $\kappa = 3 - n$ gives smaller fourth order terms of the error in (y, P_y).
- There are also other UT's available, e.g. the spherical UT which gives the same *order of* accuracy but with less sigma points ($n + 1$).

Unscented transformation

- UT actually resembles MC, but here the points are chosen in a deterministic way.
- High order information is captured using a very small amount of points.
- In UT, the distribution of x is approximated by the sigma points and the function h is kept intact. In EKF the distribution is approximated with mean and covariance, and h is approximated.
- The choice of a specific $\sqrt{P}$ does not influence the properties of UT. Cholesky factor is a good pick.

The System

We have n-state discrete-time non-linear system

$$x_{k+1} = f(x_k, u_k, t_k, w_k)$$

$$y_k = h(x_k, t_k, v_k)$$

$$w_k \sim (0, Q_k)$$

$$v_k \sim (0, R_k),$$

where we have included the noise terms in f and h since they may not be additive. If they are not additive they must be augmented to the state variable.

Unscented Kalman Filter

(0.) Initialize mean and covariance as usual
1. Do the time update from (x_{k-1}^+, P_{k-1}^+) to (x_k^-, P_k^-) using UT and f
2. (Optional) Calculate new sigma points about (x_k^-, P_k^-) (or stick with the ones from (x_{k-1}^+, P_{k-1}^+)
3. Transform the sigma points of step 2 with h to get predicted measurements sigma points $\hat{y}^{(i)}$.
4. Calculate $\hat{y}_k$, P_y, and P_{xy} as "sample statistics" of the sigma points. (note that this is not a random sample!)
5. Now, simply,

$$K_k = P_{xy} P_y^{-1}$$

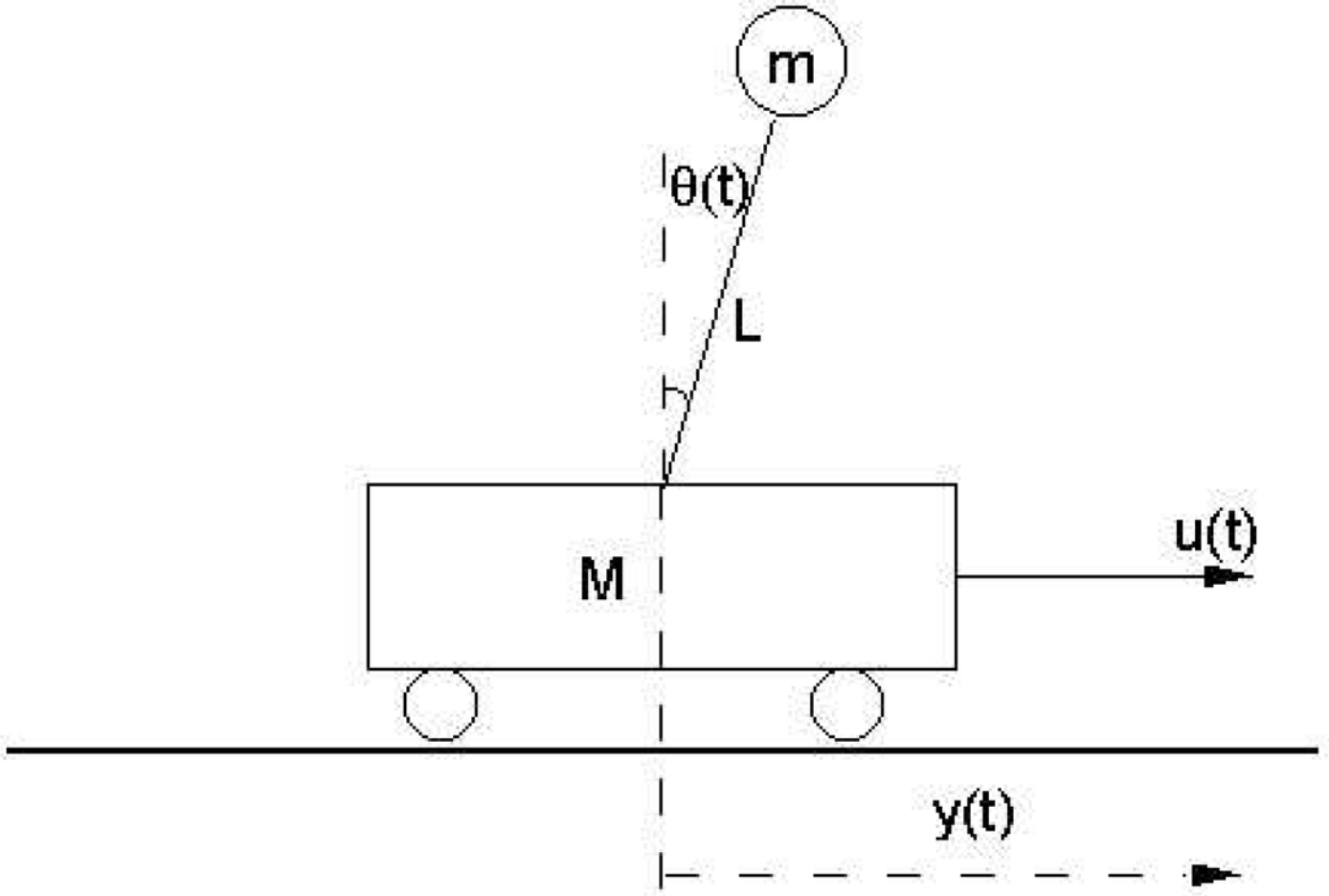
$$x_k^+ = x_k^- + K_k(y_k - \hat{y}_k)$$

$$P_k^+ = P_k - K_k P_y K_k^T$$

To UKF or not to UKF?

- Choose UKF when
  - f and/or h non-linear.
  - difficulties in EKF implementation / poor EKF performance
- Go instead for
  - KF if model is linear
  - PF if model is “strange” (f, h too non-linear, distributions are e.g. bimodal. . .)
  - EKF if it works fine and is computationally cheaper.

Example / Inverted Pendulum



Why unscented?

scented

adjective

1. having the sense of smell; "keen-scented hounds" [ant: scentless]
2. filled or impregnated with perfume; "perfumed boudoir"; "perfumed stationery"; "scented soap" [syn: perfumed]
3. having a natural fragrance; "odoriferous spices"; "the odorous air of the orchard"; "the perfumed air of June"; "scented flowers" [syn: odoriferous]
4. (used in combination) having the odor of; "clean-scented laundry"; "a manure-scented barnyard"

WordNet◯ 3.0, ◯ 2006 by Princeton University.

The following slides are adapted from Simon Julier's UKF slides:

`http://soma.crl.mcmaster.ca/ASLWeb/Resources/data/LakeLouise2003/Julier_slides_trans.pdf`

Why unscented?

- Some history:
  - The algorithm later to be known as the unscented transformation was derived in the summer of 1994 at the Robotics Research Group (RRG), Oxford UK.
  - In a 1995 ACC paper it was called the New Filter Due to the somewhat ambiguous (and boring) nature of the name, it was decided that a new and better name was needed.
- Naming process:
  - Approximately 20 different names were identified. Any involving names of authors or institutions were automatically excluded (other options included the Sigma Point Filter and the FAB Filter).
  - A democratic vote was taken by members of RRG to choose the name Unscented Kalman Filter.

But why unscented?

- Further research showed that the concept of deterministically choosing points to match statistics could be generalised out of a Kalman filter context.
- Therefore, the terminology was redefined. The UT was defined to be the transformation process to make sure that it can stand on its own.
- Any claims that the name was chosen to imply that the EKF stinks or to irritate Hugh (Durrant-Whyte) are pure speculation. Honest.
  - [Hugh Durrant-Whyte is(was?) a robotics professor. Apparenty *he* stinks. This seems to be a control theorists' inside joke]

the difficult and time-consuming homework. . .

. . . just kidding, Matlab=lotsa fun! And, moreover, ppl should have a lot of spare time during holidays. Think of it as me saving you from boredom.

Get the EKF/UKF toolbox from LCE@TKK
`http://www.lce.hut.fi/research/mm/ekfukf/`.

Do a modified version of problem 14.15 in the book. The catch: Dan's model solution is extremely messy and there are some discrepancies between it and problem statement.

the difficult and time-consuming homework. . .

The IP model and quantities are as in book. The text is modified as follows:

- The angle θ is measured every 10 ms with a std. of 0.1 rad.
  `H=[0 0 1 0];`
- Discrete-time process noise covariance is
  $Q = \text{diag}(0, 0.0004, 0, 0.04)dt$.
- Initial state is `x(:,1)=[0 0 .4 0]';` and UKF does not know it (you may init `xp(:,1)=[0 0 0 0]';`, `Pp(:,:,1)=eye(4)`)
- Feedback control is given as

$$u = 100\hat{\theta} + 20\hat{\dot{\theta}},$$

  and it is perfectly known to the UKF (see, the state is not known but the control is, which is somewhat unrealistic assumption).

the difficult and time-consuming homework. . .

Since I appreciate being alive and all, here's some hints:

- Do not look at Dan's code pp. 285→ in the the solution manual. It will probably just confuse you.
- If you disregarded previous hint: You'll be better off by defining only an 4×N array `x` for the state instead of Dan's 10+ variables on the order of `theta`, `thetadot`, `thetadotdot_array`, `theta_old`, and so on. At least I had a near-death experience with those.
- Get my supplementary functions from `http://www.tkk.fi/~pjanis/`. Or write your own if you don't trust me. (Might even be wise. . .)
- Take a look at the toolbox manual
- If all else fails, email `pekka.janis@tkk.fi` for advice.